

Attack Map

Security Posture

Overall Risk Score

72 / 100

Medium Risk

Threats Detected

Last 24 Hours

1,248

Top Threat Categories

Last 24 Hours



Malware	42%
Brute Force	24%
Exploit	18%
Reconnaissance	10%
Other	6%

Recent Security Alerts

Time	Severity	Alert Name
10:24:15	High	Possible Lateral Movement
10:18:42	High	Suspicious PowerShell Activity
10:14:09	Medium	Possible Credential Dumping
10:07:33	Medium	Unusual Network Connection
10:02:11	Low	System Service Discovery

REAL-TIME SOC THREAT DETECTION & INCIDENT RESPONSE LAB

Enterprise Security Monitoring using Splunk SIEM



PREPARED BY

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Aspiring SOC Analyst | Cybersecurity Enthusiast |
Threat Detection & Incident Response

TECHNOLOGY STACK

- Splunk Enterprise SIEM
- Windows 10
- Sysmon
- Splunk Universal Forwarder
- Kali Linux
- VirtualBox / VMware
- MITRE ATT&CK Framework
- Detection Engineering (SPL Queries)

PROJECT FOCUS

- Real-Time Security Monitoring
- Threat Detection
- Incident Investigation
- Alert Triage
- Threat Hunting
- SOC Workflow Simulation

LINKS



GitHub

<https://github.com/parag1020>



LinkedIn

<https://www.linkedin.com/in/parag-patel-9593742a7>



VERSION
Version 1.0



YEAR
2026

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1. Project Overview

Introduction

Modern organizations generate massive volumes of security logs from endpoints, servers, network devices, and applications every second. Monitoring these events manually is inefficient, time-consuming, and often leads to delayed threat detection.

Security Operations Centers (SOC) rely on Security Information and Event Management (SIEM) platforms to collect, normalize, analyse, and monitor security events in real-time. These platforms help analysts identify suspicious behavior, investigate incidents, and respond quickly to cyber threats.

The purpose of this project is to design and implement a practical real-time SOC monitoring lab using Splunk Enterprise SIEM. This environment simulates enterprise-style security monitoring by integrating endpoint telemetry, centralized log collection, attack simulation, and threat investigation workflows.

The lab consists of three primary components: a Kali Linux attacker machine used to simulate adversarial activity, a Windows 10 monitored endpoint configured with Sysmon for enhanced telemetry, and a Splunk Enterprise server responsible for centralized log analysis and alert monitoring.

The project focuses on simulating realistic attack scenarios commonly encountered in modern enterprise environments, including reconnaissance scanning, brute-force authentication attempts, suspicious PowerShell activity, and abnormal process execution. These attack simulations generate security events that are collected and analysed within Splunk to demonstrate real-time detection capability.

This implementation is designed to replicate practical SOC analyst responsibilities such as alert triage, incident investigation, detection engineering, and threat hunting. Rather than focusing only on theoretical cybersecurity concepts, this project emphasizes operational hands-on monitoring workflows like real-world SOC environments.

By completing this project, practical skills in SIEM monitoring, Windows event analysis, endpoint telemetry interpretation, Splunk query development, incident analysis, and threat detection engineering are demonstrated in a recruiter-focused technical portfolio format.

2. Project Objectives

Primary Objective

The primary objective of this project is to design and implement a practical real-time Security Operations Center (SOC) monitoring environment capable of detecting, analysing, and investigating simulated cyber threats using Splunk Enterprise SIEM.

Specific Objectives

This project is designed to achieve the following technical objectives:

- Build a centralized log monitoring environment using Splunk Enterprise.
- Configure Windows endpoint monitoring for security event collection.
- Deploy Sysmon for enhanced endpoint telemetry visibility.
- Configure Splunk Universal Forwarder for secure log ingestion.
- Simulate realistic cyber-attack scenarios using Kali Linux.
- Detect reconnaissance activities such as port scanning.
- Monitor authentication abuse including brute-force attempts.
- Identify suspicious PowerShell execution activity.
- Analyse process creation events and abnormal endpoint behavior.
- Develop Splunk SPL detection queries for security monitoring.
- Create operational SOC dashboards for real-time visibility.
- Perform incident investigation using collected telemetry.
- Practice threat hunting workflows using SIEM search queries.
- Map observed activity to MITRE ATT&CK techniques.
- Document the project as a professional cybersecurity portfolio case study.

Expected Learning Outcomes

By completing this project, the following practical cybersecurity capabilities are demonstrated:

- SIEM monitoring
- Alert triage
- Incident investigation
- Log analysis
- Endpoint telemetry interpretation
- Threat detection engineering
- Threat hunting
- Security dashboard creation
- SOC workflow understanding
- Practical enterprise-style monitoring experience
- Security event correlation
- Endpoint threat visibility
- Detection rule creation
- Attack behavior analysis
- Practical SIEM investigation skills

3. Architecture Design

SOC Lab Architecture

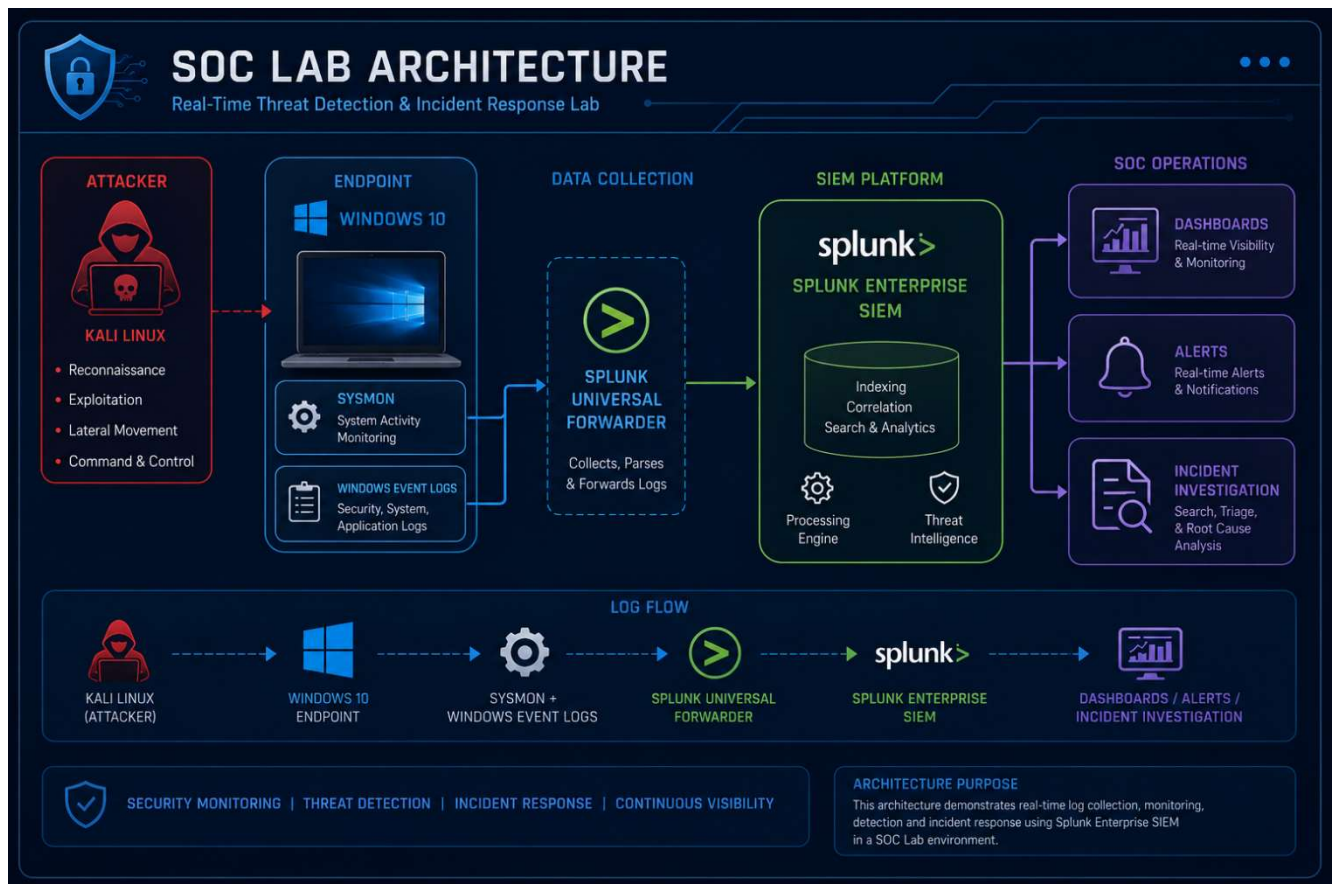


Figure 1: Enterprise-style architecture of the Real-Time SOC Threat Detection & Incident Response Lab.

4. System Requirements

Host Machine Requirements

This SOC lab requires sufficient hardware resources to run multiple virtual machines, Splunk SIEM, endpoint monitoring, and attack simulations efficiently.

Minimum Requirements

Component	Minimum Specification
Processor	Intel Core i5 / AMD Ryzen 5
RAM	16 GB
Storage	100 GB Free SSD Space
Virtualization	Intel VT-x / AMD-V Enabled
Operating System	Windows 10 / Windows 11

Recommended Requirements

Component	Recommended Specification
Processor	Intel Core i7 / Ryzen 7
RAM	32 GB
Storage	250 GB SSD
Virtualization	Enabled
Display	Full HD Recommended

Virtual Machine Allocation

Virtual Machine	RAM	CPU	Storage
Windows 10 Endpoint	4 GB	2 Cores	60 GB
Kali Linux Attacker	4 GB	2 Cores	40 GB
Splunk Enterprise Server	8 GB	4 Cores	80 GB

5. Software & Downloads

Required Software Components

This section lists all software tools, operating system images, and supporting components required to build the Real-Time SOC Threat Detection Lab. These tools are selected to simulate a practical enterprise security monitoring environment, including endpoint telemetry collection, SIEM log analysis, attack simulation, and incident investigation.

Core Software Components

Software	Purpose
Splunk Enterprise	SIEM platform for log analysis, dashboards, and alert monitoring
Splunk Universal Forwarder	Log forwarding agent for Windows endpoint
Sysmon	Advanced Windows system activity monitoring
Kali Linux	Attack simulation platform
Windows 10 ISO	Monitored target endpoint
VirtualBox / VMware Workstation	Virtual machine hosting platform
Google Chrome / Firefox	Access Splunk web interface
7-Zip	File extraction utility

Download Sources

- **Splunk Enterprise — splunk.com**
- **Splunk Universal Forwarder — splunk.com**
- **Sysmon — [Microsoft Sysinternals](https://www.microsoft.com/sysinternals)**
- **Kali Linux — kali.org**
- **Windows 10 ISO — [Microsoft](https://www.microsoft.com)**
- **VirtualBox — [virtualbox.org](https://www.virtualbox.org)**
- **VMware Workstation — [vmware.com](https://www.vmware.com)**

Before installation, verify that all required software packages are downloaded from official vendor sources to maintain integrity and security. Using authentic installation packages reduces compatibility issues and minimizes the risk of introducing malicious or tampered software into the SOC lab environment.

6. Implementation and Lab Setup

6.1 Virtualization Environment Setup

To simulate a real-world Security Operations Center (SOC) environment safely, the entire lab was deployed using virtualization technology. This approach allows multiple operating systems to run simultaneously on a single host machine while maintaining isolation, flexibility, and controlled experimentation.

A dedicated virtual lab environment was created to replicate enterprise-style infrastructure for security monitoring, attack simulation, and incident investigation.

The virtualization environment included three primary systems:

- Windows 10 Endpoint Machine
- Kali Linux Attacker Machine
- Splunk Enterprise SIEM Server

Virtualization platforms such as VirtualBox or VMware Workstation were used to provision and manage these systems.

A private virtual network was configured to allow secure communication between all virtual machines.

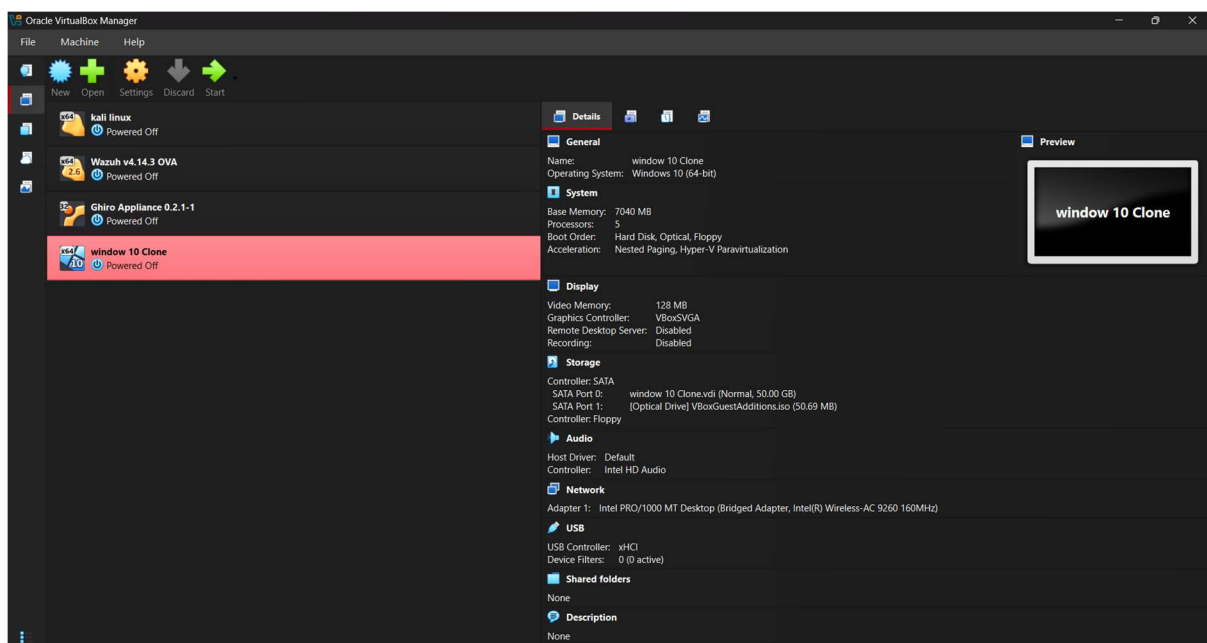


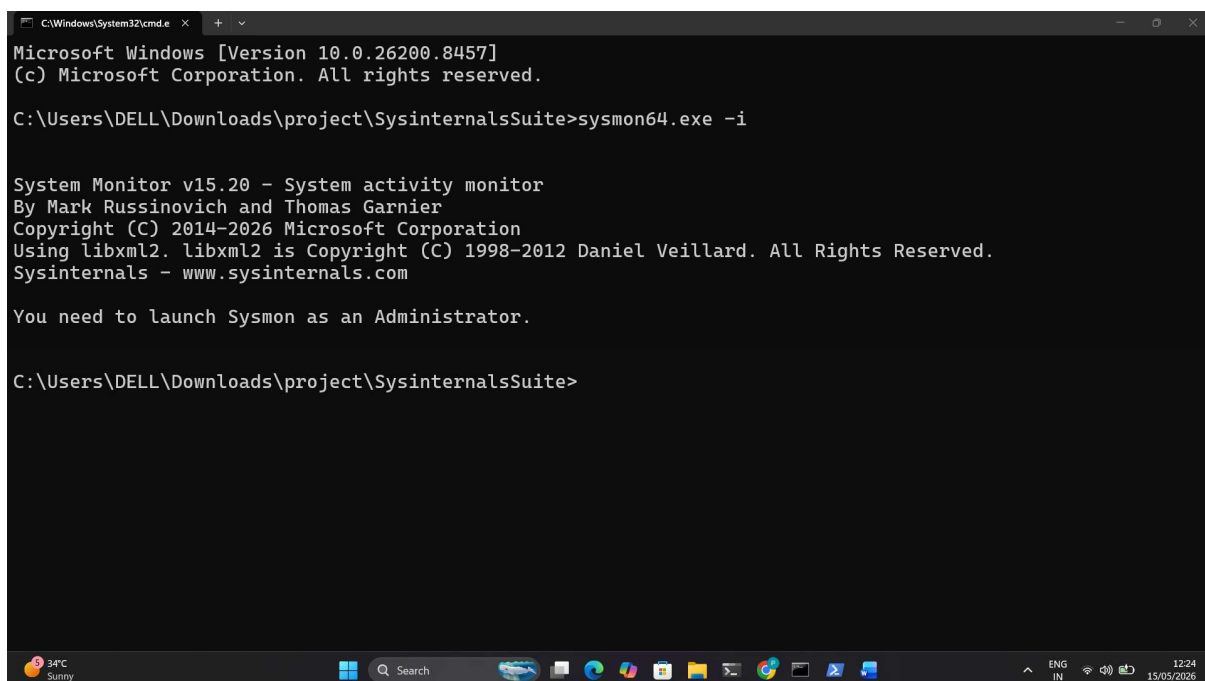
Figure 2: Virtualized SOC Lab Environment in Oracle VirtualBox.

6.2 Endpoint Monitoring Setup

The Windows 10 endpoint machine was configured for advanced system activity monitoring using Sysmon. Sysmon provides detailed telemetry such as process creation, network connections, file modifications, and registry activity, which are essential for threat detection.

To forward endpoint logs into the SIEM platform, Splunk Universal Forwarder was installed and configured on the Windows endpoint. This enabled real-time transmission of Windows Event Logs and Sysmon telemetry to Splunk Enterprise for centralized analysis.

The endpoint monitoring setup improved visibility into endpoint behavior and formed the primary data source for attack detection and incident investigation.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL\Downloads\project\SysinternalsSuite>sysmon64.exe -i

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

You need to launch Sysmon as an Administrator.

C:\Users\DELL\Downloads\project\SysinternalsSuite>
```

Figure 3: Sysmon Successfully Installed and Running on Windows Endpoint

6.3 Log Collection and SIEM Integration

Splunk Enterprise was configured to enable centralized log collection within the SOC lab environment. A forwarding host was successfully configured using TCP port 9997, establishing the communication channel required for endpoint log transmission.

This forwarding configuration allows Windows endpoint telemetry, security logs, and monitored system activity to be transmitted to the Splunk SIEM platform for analysis and correlation.

The successful forwarding setup forms the core of the log collection architecture, enabling centralized monitoring, alert generation, and incident investigation workflows in the simulated SOC environment.

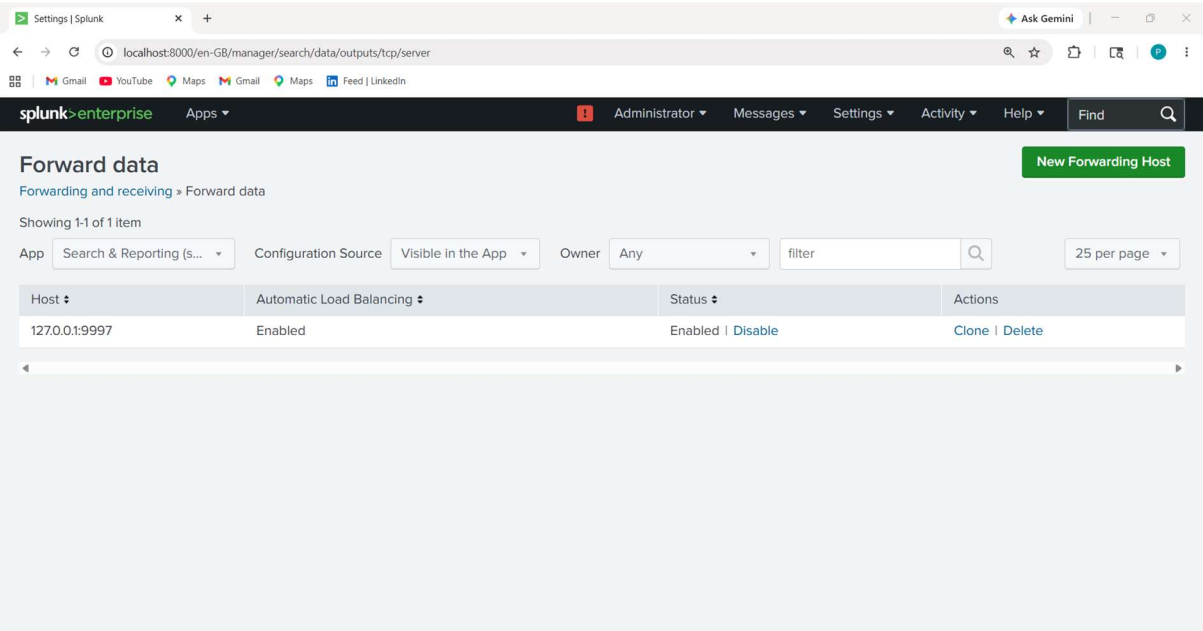


Figure 4: Splunk Forwarding Host Configuration for Centralized SIEM Log Collection

6.4 Splunk Data Input Configuration

Splunk Enterprise was configured to collect multiple categories of Windows telemetry required for effective SOC monitoring. Data input configuration included Windows Security logs, System logs, Application logs, and Sysmon operational logs to provide comprehensive endpoint visibility.

These data sources enable monitoring of authentication events, process execution, system changes, network activity, and administrative actions occurring on the monitored Windows endpoint. Proper input configuration ensures that critical events are continuously indexed and available for security analysis.

The configured data collection framework strengthens the SIEM environment by establishing a centralized telemetry pipeline capable of supporting threat detection, alert generation, and incident investigation workflows.

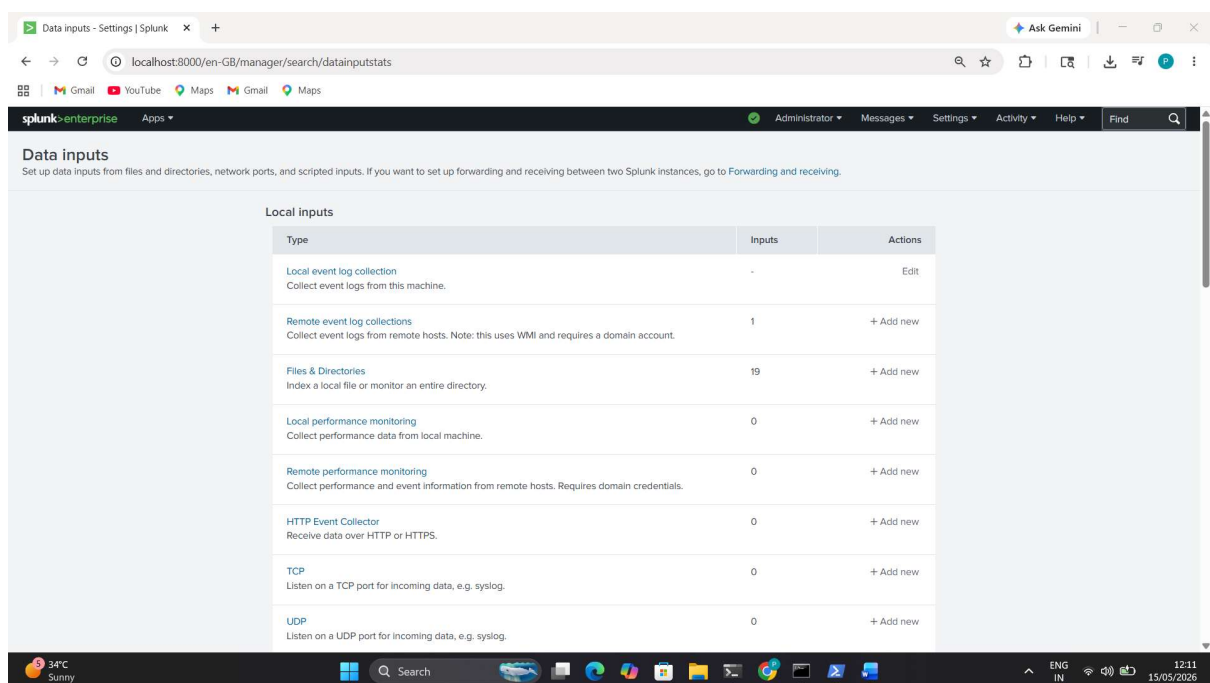


Figure 5: Splunk Data Input Configuration for Windows Event Log and Endpoint Telemetry Collection

6.5 Detection Query Development

Custom detection logic was developed within Splunk Enterprise using Search Processing Language (SPL) queries to identify suspicious activity and validate successful telemetry ingestion from the monitored Windows endpoint. These detection queries simulate the workflow of a real Security Operations Center (SOC), where analysts continuously query, filter, and investigate system events for indicators of malicious behavior.

The developed queries were designed to monitor authentication failures, suspicious process execution, PowerShell activity, administrative commands, and endpoint telemetry generated by Windows Event Logs and Sysmon. By leveraging Splunk's search and correlation capabilities, relevant security events can be rapidly identified and investigated.

A general validation query was first executed to confirm that event logs were successfully being indexed within Splunk. This verification established that the configured data sources were functioning correctly and that the SIEM platform was receiving real-time endpoint telemetry.

The implementation of custom detection queries significantly enhances monitoring capability, improves visibility into system behavior, and forms the analytical foundation for attack detection and incident response within the SOC lab environment.

Validation SPL Query Used:

```
index=* | head 20
```

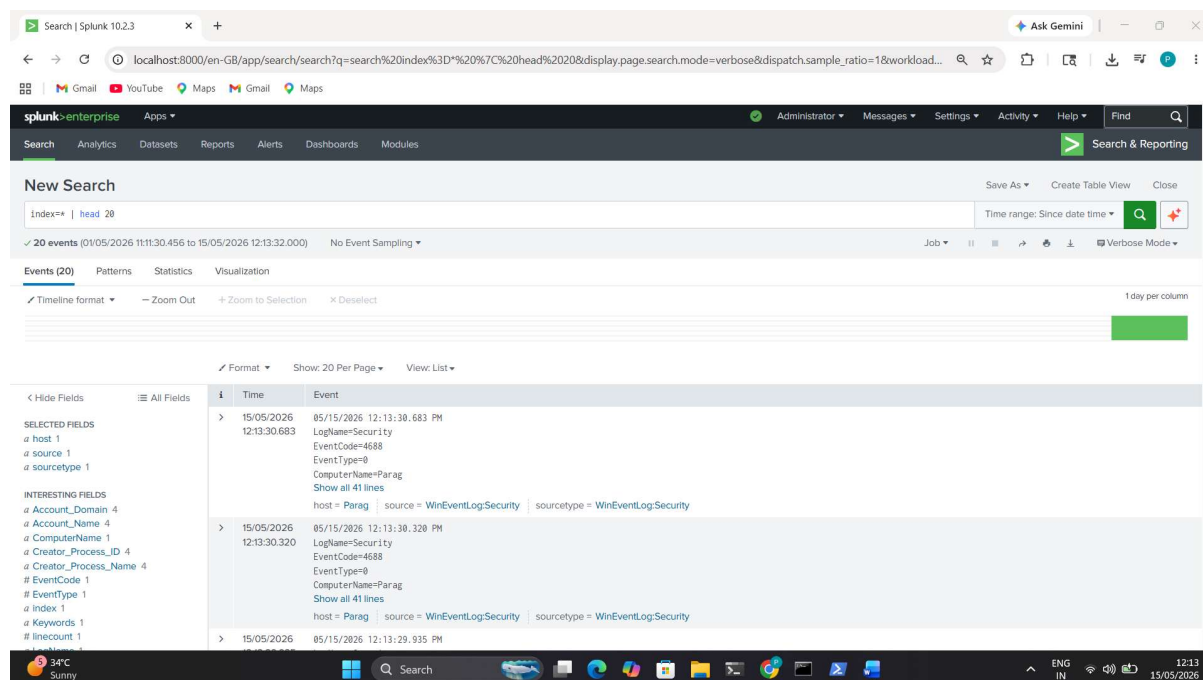


Figure 6: Custom SPL Detection Query Validation within Splunk Enterprise SIEM

6.6 Baseline Endpoint Activity Monitoring and Telemetry Validation

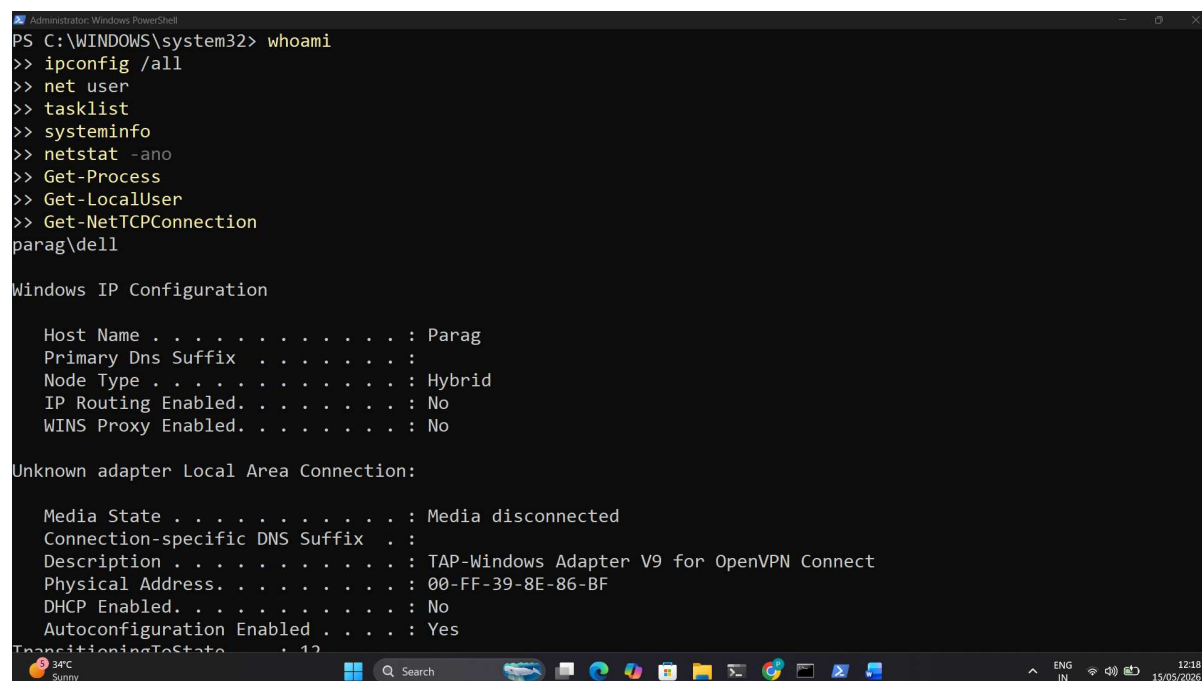
To validate successful telemetry collection within the SOC environment, baseline administrative activity was intentionally generated on the monitored Windows endpoint. This process helps confirm that Windows Event Logs, Sysmon telemetry, and Splunk log forwarding are functioning correctly.

A set of common administrative and system information commands were executed using elevated PowerShell and Command Prompt sessions to simulate normal enterprise administrator behavior. These actions generated security, process, network, and system activity logs that were successfully ingested into Splunk Enterprise for monitoring and analysis.

Establishing baseline activity provides a trusted reference for distinguishing legitimate administrative actions from suspicious behavior during later attack simulations.

Administrative Commands Executed:

```
whoami
ipconfig /all
net user
tasklist
systeminfo
netstat -ano
Get-Process
Get-LocalUser
Get-NetTCPConnection
```



```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> whoami
>> ipconfig /all
>> net user
>> tasklist
>> systeminfo
>> netstat -ano
>> Get-Process
>> Get-LocalUser
>> Get-NetTCPConnection
parag\de11

Windows IP Configuration

Host Name . . . . . : Parag
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Unknown adapter Local Area Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9 for OpenVPN Connect
Physical Address. . . . . : 00-FF-39-8E-86-BF
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
TransitioningToState . . . . . : 12
```

Figure 7: Baseline Administrative Activity Generated for Endpoint Telemetry Validation

7. Attack Simulation and Threat Detection

7.1 Attack Simulation Overview

To validate the effectiveness of the SOC monitoring environment, controlled cyber-attack simulations were performed against the monitored Windows endpoint. These simulations were designed to generate realistic attacker behavior and validate whether the deployed monitoring infrastructure could successfully capture and analyse suspicious activity.

The attack simulation focused on two common real-world attack scenarios: reconnaissance scanning and authentication abuse. These activities generated observable telemetry within the monitored environment, which was then analysed using Splunk Enterprise SIEM.

This practical phase demonstrates the ability of the SOC lab to simulate attacker activity, monitor security events, and support analyst-driven detection workflows.

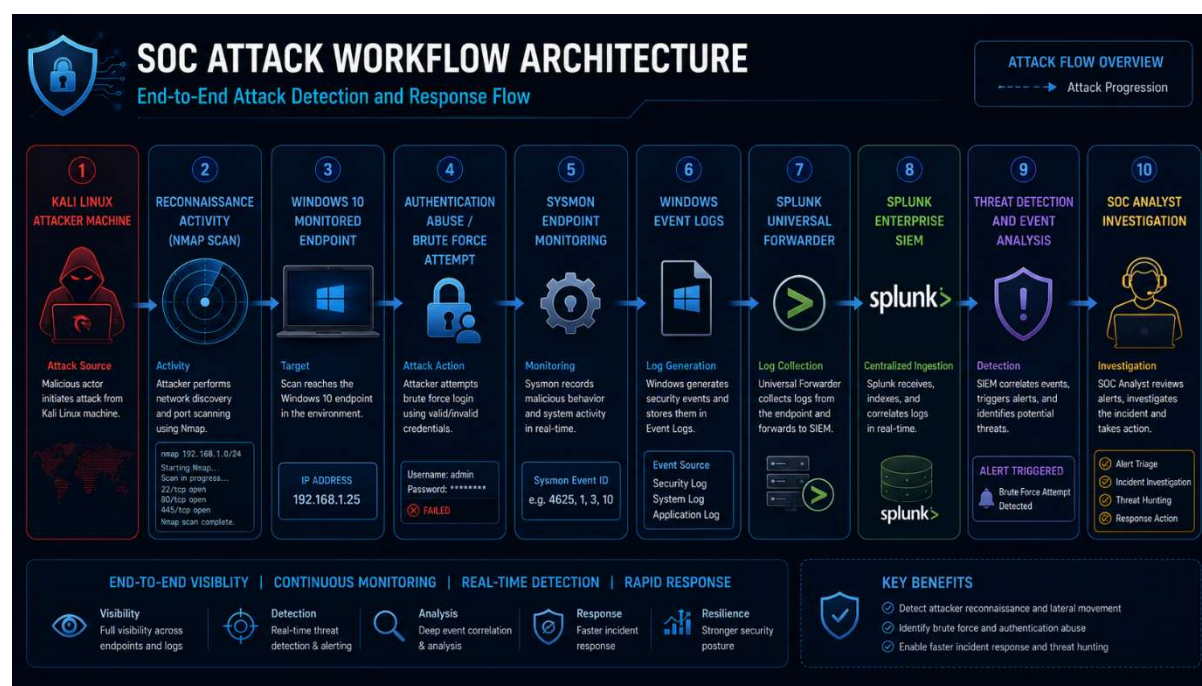


Figure 8: SOC Attack Simulation and Threat Detection Workflow

7.2 Reconnaissance Activity Simulation (Nmap Scan)

Reconnaissance is one of the earliest stages of a cyber intrusion, where attackers attempt to identify active hosts, open ports, and exposed network services before launching further attacks.

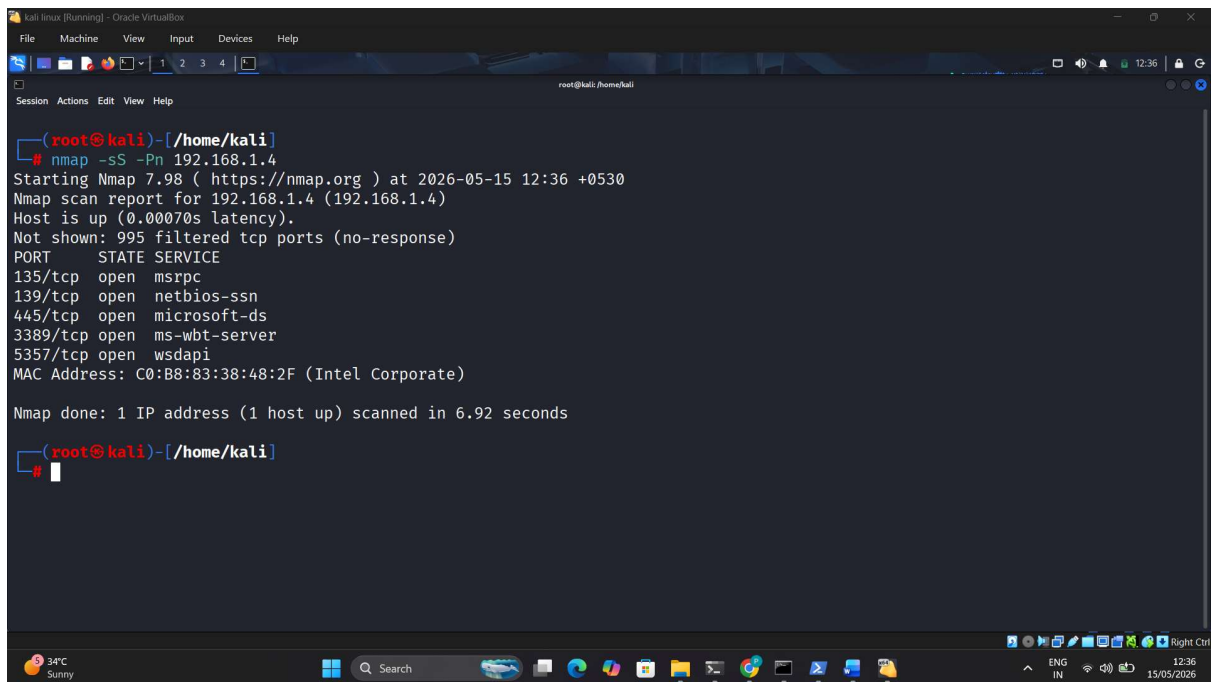
To simulate realistic attacker reconnaissance behavior, an Nmap SYN scan was executed from the Kali Linux attacker machine against the monitored Windows endpoint within the SOC lab environment.

The following command was used:

```
nmap -sS -Pn 192.168.1.4
```

The scan successfully identified the target host as active, demonstrating realistic attacker discovery behavior commonly observed during the early phases of cyber-attacks.

This simulation validates the SOC lab's ability to represent reconnaissance activity and generate observable security telemetry for monitoring and investigation.



```
(root@kali)-[/home/kali]
# nmap -sS -Pn 192.168.1.4
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-15 12:36 +0530
Nmap scan report for 192.168.1.4 (192.168.1.4)
Host is up (0.00070s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsddapi
MAC Address: C0:B8:83:38:48:2F (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 6.92 seconds

(root@kali)-[/home/kali]
```

Figure 9: Simulated Network Reconnaissance Activity Using Nmap Scan

7.3 Authentication Abuse Simulation (Brute Force Activity)

Authentication abuse is a common attack technique in which threat actors repeatedly attempt invalid login requests to gain unauthorized access to target systems.

To simulate brute force authentication behavior, repeated failed login attempts were intentionally generated against the monitored Windows system within the SOC lab environment.

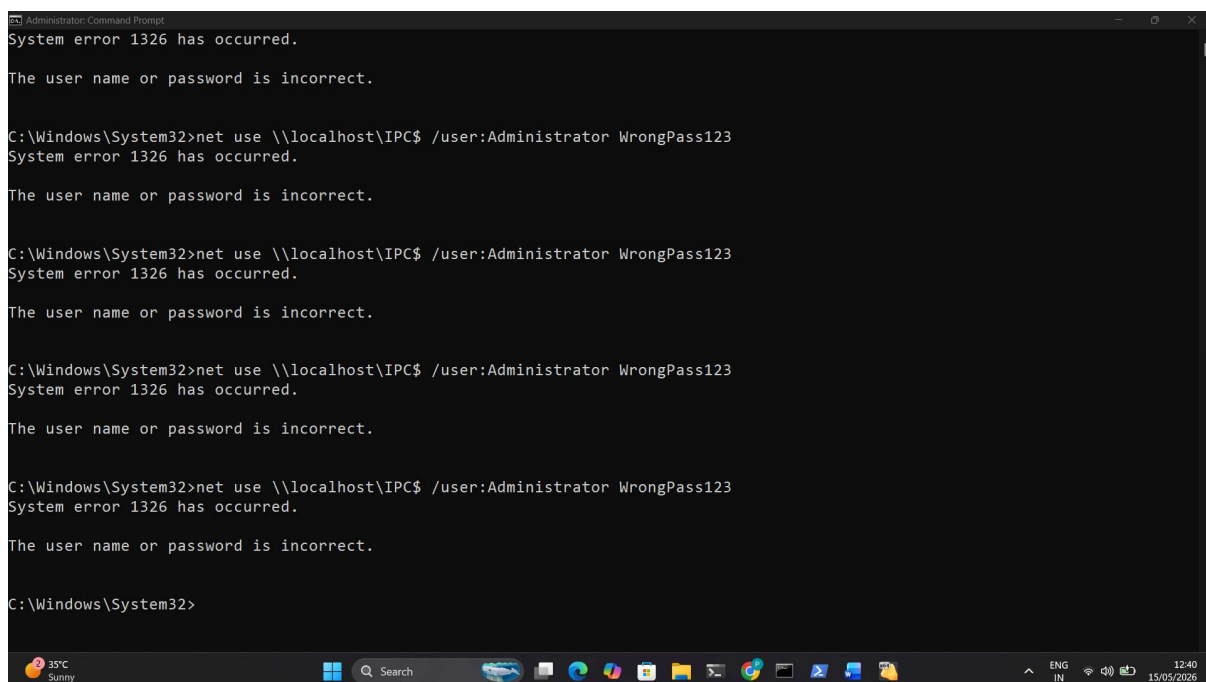
The following command was executed multiple times:

```
net use \\localhost\IPC$ /user:Administrator WrongPass123
```

This simulation generated failed authentication activity representing password guessing behavior commonly associated with brute force attacks.

Such events are important detection use cases for Security Operations Centers, as repeated failed authentication attempts may indicate credential abuse, unauthorized access attempts, or malicious user activity.

This controlled simulation demonstrates how authentication abuse scenarios can be represented within a practical SIEM monitoring environment.



```
Administrator: Command Prompt
System error 1326 has occurred.
The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.
The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.
The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.
The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.
The user name or password is incorrect.

C:\Windows\System32>
```

Figure 10: Simulated Brute Force Authentication Abuse Activity Against Windows Endpoint

The generated failed authentication attempts were successfully detected within Splunk Enterprise SIEM through Windows Security Event monitoring. Multiple failed logon events (Event ID 4625) were observed, confirming successful telemetry collection and detection visibility for authentication abuse activity.

This detection validates the effectiveness of the deployed SOC monitoring environment in identifying brute force-style login failures and suspicious authentication behavior.

Query Used:

```
index=* EventCode=4625 | table _time, Account_Name, Workstation_Name, Failure_Reason
```

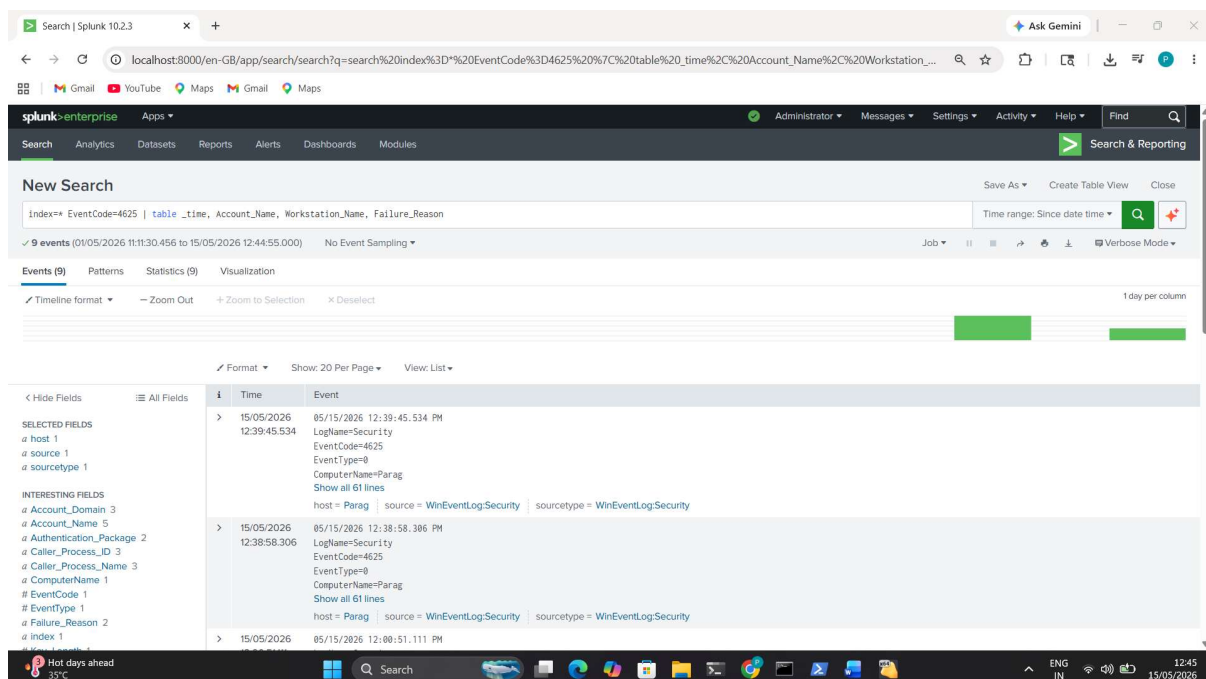


Figure 11: Splunk Detection of Failed Authentication Attempts During Brute Force Simulation

7.4 Event Visibility in Splunk SIEM

Following the execution of simulated attack activities, Splunk Enterprise SIEM was used to verify successful log ingestion and event visibility from the monitored Windows endpoint.

A validation search query was executed to confirm that generated security telemetry was successfully indexed within the SIEM platform.

Query used:

index=* | head 20

The search results confirmed successful event ingestion, including Windows Security logs, process activity, and monitored endpoint telemetry. This validated that the deployed monitoring architecture was functioning correctly and capable of supporting security investigations.

This step confirms the operational readiness of the SOC monitoring environment for real-time threat analysis.

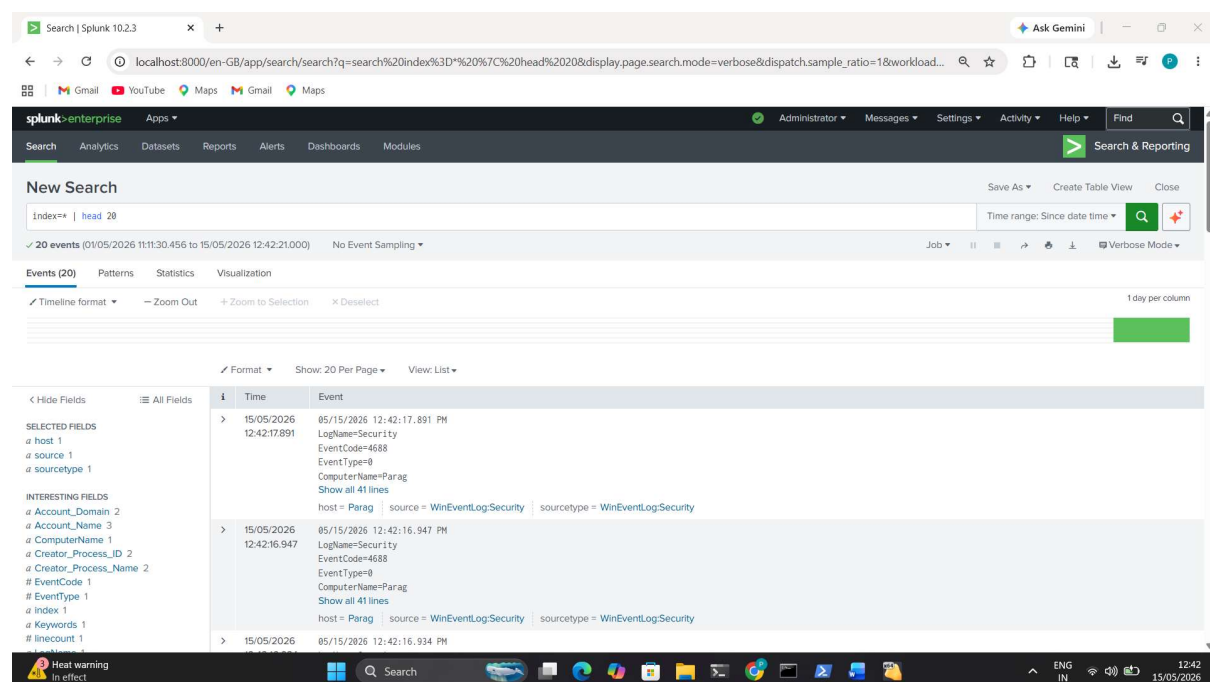


Figure 12: Successful Event Visibility and Log Ingestion in Splunk Enterprise

8. Results and Findings

The implementation of the SOC monitoring environment successfully demonstrated practical endpoint monitoring, centralized log collection, attack simulation visibility, and SIEM-based threat analysis within a controlled lab environment.

The deployed architecture successfully integrated Windows endpoint telemetry, Sysmon monitoring, Splunk Enterprise SIEM, and simulated attack behavior for practical security monitoring validation.

The major outcomes observed during project implementation include:

- Successful deployment of centralized Security Information and Event Management (SIEM) infrastructure
- Successful Windows endpoint telemetry monitoring using Sysmon
- Successful log ingestion and event indexing within Splunk Enterprise
- Detection visibility for simulated reconnaissance activity using Nmap
- Detection visibility for brute force authentication abuse using failed login event monitoring
- Successful monitoring of Windows Security Event telemetry
- Practical validation of Splunk Search Processing Language (SPL) based threat investigation
- Improved practical understanding of SOC analyst monitoring workflows and incident visibility processes

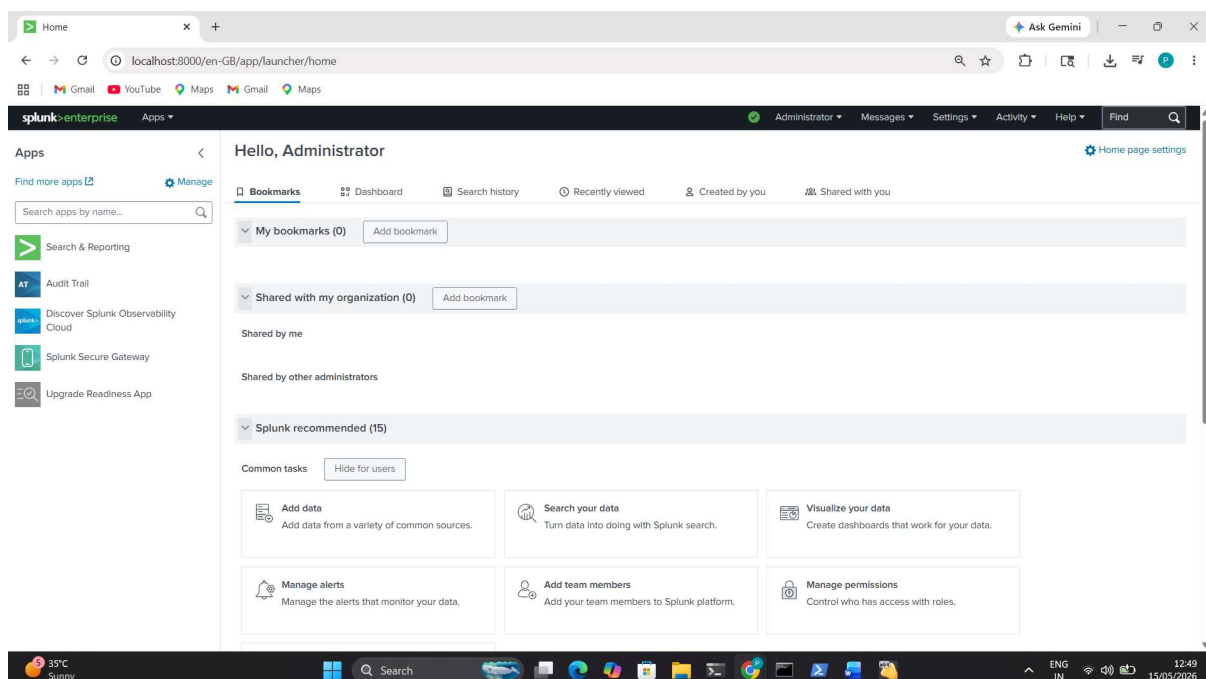


Figure 13: Splunk Enterprise SIEM Dashboard Showing Successful SOC Monitoring Environment

9. Conclusion

This project successfully demonstrated the design and implementation of a Security Operations Center (SOC) monitoring environment using Splunk Enterprise and Windows endpoint log collection.

The project included endpoint monitoring setup, centralized log collection, SIEM integration, reconnaissance simulation using Nmap, and brute force attack simulation through failed authentication attempts. These activities generated real security events, which were successfully collected, indexed, and analysed in Splunk.

Through event correlation and log analysis, suspicious activities such as unauthorized login attempts, process execution events, and network-related system behavior were identified effectively. The project highlighted the practical importance of centralized monitoring, real-time event visibility, and security analytics in detecting potential cyber threats.

Overall, this SOC implementation provided hands-on experience with SIEM operations, attack monitoring, and incident analysis in a simulated enterprise security environment. The project demonstrates how SOC technologies can improve threat detection, visibility, and incident response capabilities in modern cybersecurity infrastructure.